



WORKSHOP

Critical Infrastructures and Accessibility Analysis under Flood Conditions



Flood
hazard



Critical
infrastructure



Accessibility
analysis



Societal
impacts



Thursday 8th October 2026



Karlstad, Sweden



Workshop Facilitator:

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**Workshop
GitHub Repository**

Scan the QR code or click the link to access
all workshop notebooks, datasets and
supporting files.



https://github.com/amaletaa/SESAC_RisCon26_Workshop





Turning Global Pixels into Social Intelligence



Competence

Lowering the barrier to
satellite-based analysis.

- Open, reusable workflows & code
- Training and short courses for social scientists
- Knowledge hub for Copernicus data in Sweden



Collaboration

Bridging academia, public
authorities and industry.

- Stakeholder-driven research questions
- Co-creative hackathons & workshops
- Industry collaborations
- Academic partnerships across Sweden



Innovation

Pushing the technical
frontier of GeoAI.

- Geospatial & deep learning on imagery
- Fusing satellite data with population registers
- Decision-support tools for public policy



Our events - Webinars,
Seminars, Hackathons



Community survey to
shape our future
activities

Critical Infrastructure Facilities

“Systems, facilities, and assets that are vital for the functioning of society & economy”



Healthcare

Hospitals, clinics



Emergency

Fire, police, rescue



Education

Schools, universities



Flooding: A Growing Global Challenge



One of the most frequent natural hazards worldwide



€12B average annual flood losses in the EU



Disruption of the road network & critical facilities operation



Increasing frequency & intensity due to climate change



1. Water Depth (d)

- Maximum or average depth of floodwater
- Determines the extent of submergence and damage potential



2. Water Velocity (v)

- Speed of floodwater flow
- Influences the destructive force, erosion and stability of structures



3. Flood Duration (T)

- Length of time the area remains flooded
- Affects damage accumulation, recovery time and service disruption



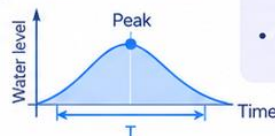
4. Flood Extent (A)

- Spatial coverage of flooding (inundated area)
- Determines the number of elements at risk and impacts on networks



5. Rate of Rise / Time to Peak

- How quickly water levels rise and reach their peak
- Critical for preparedness and early warning



Flood Scenarios

Returning Period (T_R)

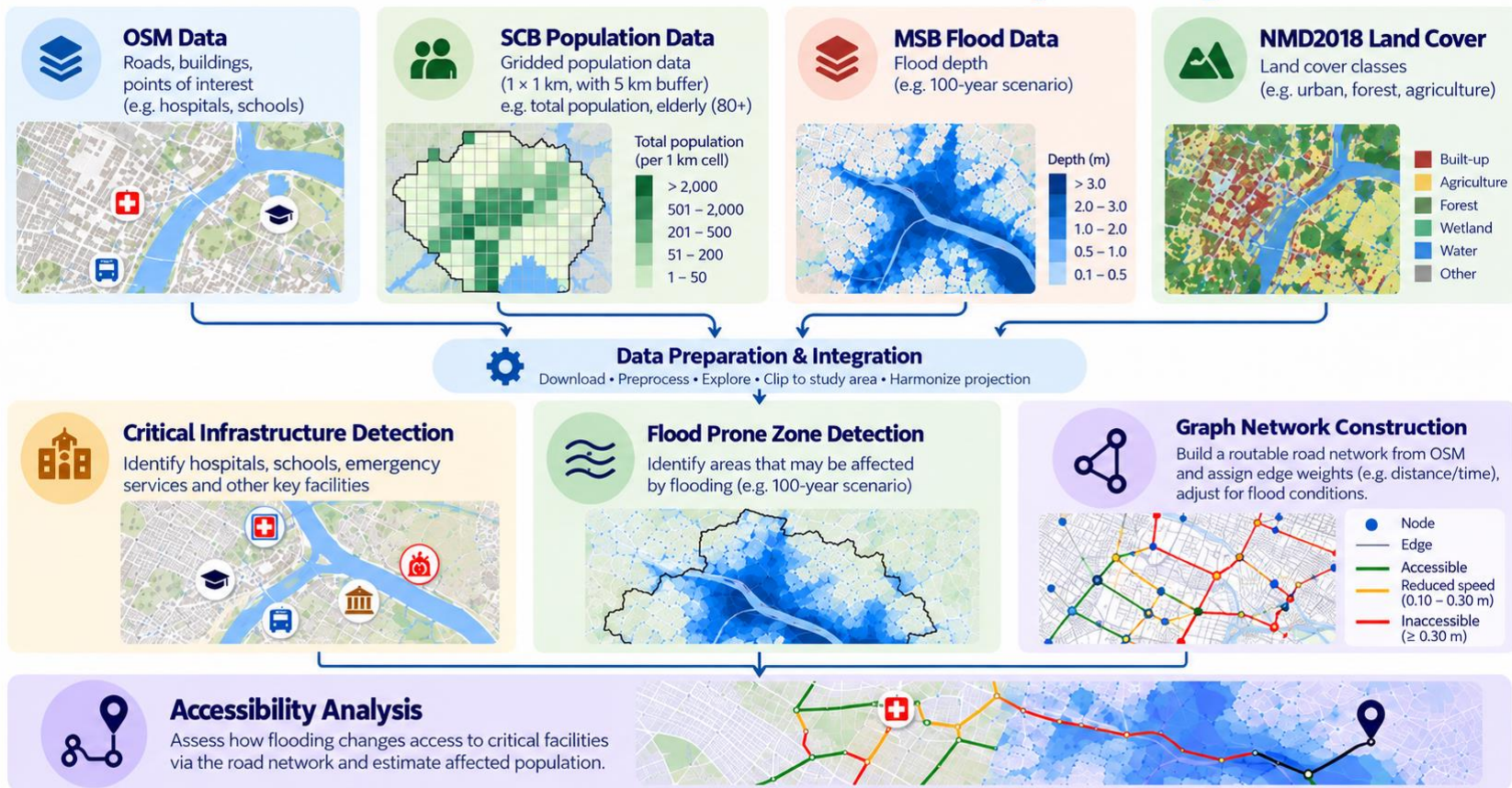
Average time between events of similar magnitude.

Returning Period T_R (years)	Annual Exceedance Probability (AEP)
10	10%
20	5%
50	2%
100	1%
200	0.5%
500	0.2%
1000	0.1%



A flood with $T_R = 100$ years has a 1% chance of being exceeded in any given year.

Workflow: From Data to Accessibility Analysis



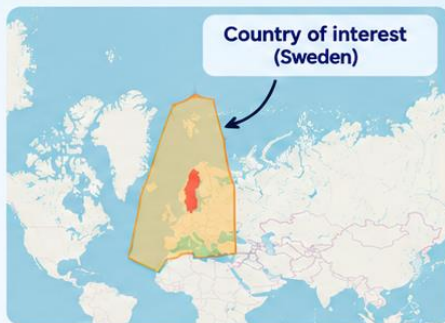
OpenStreetMap (OSM) Dataset



OpenStreetMap

A free, open and collaborative geospatial database

OpenStreetMap (OSM) provides global, up-to-date geographic data created and maintained by a large community of contributors. It includes detailed information on roads, buildings, points of interest, land use and many other features, and is freely available for research and non-commercial use.



Download OSM data (Geofabrik)

<https://download.geofabrik.de/europe/sweden.html>



Data source
(Geofabrik)

Sweden

Sub Regions

Name	Quick Links		Coordinates
Sweden	[.osm.pbf]	[.shp.zip]	56.5464 / 16.5467
Blekinge	[.osm.pbf]	[.shp.zip]	56.2 / 15.5
Dalarna	[.osm.pbf]	[.shp.zip]	60.7 / 15.0
Gävleborg	[.osm.pbf]	[.shp.zip]	61.0 / 17.0
Gotland	[.osm.pbf]	[.shp.zip]	57.5 / 18.5
Halland	[.osm.pbf]	[.shp.zip]	57.0 / 12.5
Jämtland	[.osm.pbf]	[.shp.zip]	63.5 / 13.5
...	...	...	...

Download .osm.pbf file
for Sweden
(~ 800 MB)



We will use the .osm.pbf file for Sweden and extract the data for our study area (e.g. Karlstad).



Key characteristics



Points of Interest (POIs)

Hospitals, schools, fire stations, police stations, substations, restaurants, etc.



Lines

Road network, railways, rivers, power lines, pipelines, footpaths, etc.



Polygons

Building footprints, land use, parks, lakes, airports, industrial areas, administrative boundaries, etc.



Attributes (tags)

Rich metadata describing each feature (e.g. highway, building, amenity, power, healthcare).



Example: Extracting OSM features with Python

```
# 1. Hospitals (amenity = hospital or healthcare tagging)
hospitals_all = ox.features_from_place(place, {
    "amenity": "hospital"
})
```

```
# 2. Fire stations
fire_stations_all = ox.features_from_place(place, {
    "amenity": "fire_station"
})
```

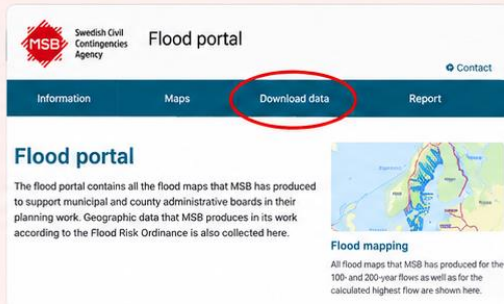
MSB Flood Data

Flood hazard maps for river flooding in Sweden



Swedish Civil
Contingencies
Agency

The MSB flood portal provides flood hazard maps for Swedish rivers, produced to support municipal and county risk planning. Maps are available for different return periods and can be downloaded for further analysis.



Data format and content



GeoTIFF (.tif)
Raster format
(~ 800 MB per file)



Flood depth
Water depth
(in meters)



Multiple scenarios
e.g. 50-, 100-, 200-year
return periods



High resolution
Typically 2 m x 2 m
grid cells



Key hazard information

- **Flood extent** – spatial coverage of the inundated area.
- **Water depth** – depth of flooding (m).
- **Water velocity** – flow velocity (m/s) (available for some areas).
- **Multiple return periods** – e.g. 50-, 100-, 200-year scenarios.

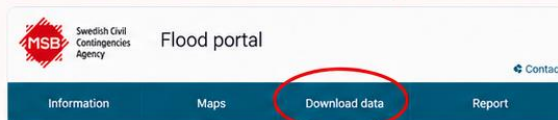


Steps to manually download flood files

1

Go to the MSB flood portal

<https://gisapp.msb.se/Apps/oversvamningsportal/index.html>



2

Choose the data format

Select raster datasets (GeoTIFF) for flood depth (and velocity if available).

In what format do you want to download the data?

☒ **Download data in GIS layers**
For use and processing in GIS
software that can handle these or
similar files.

☐ **Use a Web Map Service (WMS)**
For presentation in GIS software.
Applicable for low resolution or
application that can handle WMS.

☐ **Hydraulic models**
For your own hydraulic modelling,
please note that HEC-RAS or MIKE21 is
needed.

3

Define the area

Select county and municipality (e.g. Värmland County → Karlstad Municipality).

Värmland County

Karlstad Municipality

4

Download the data

Choose the desired flood scenario (e.g. 100-year flood) and download the files.

Flood mapping

- ☒ Flood depth (raster, GeoTIFF)
- ☒ Flood velocity (raster, GeoTIFF)



Download data

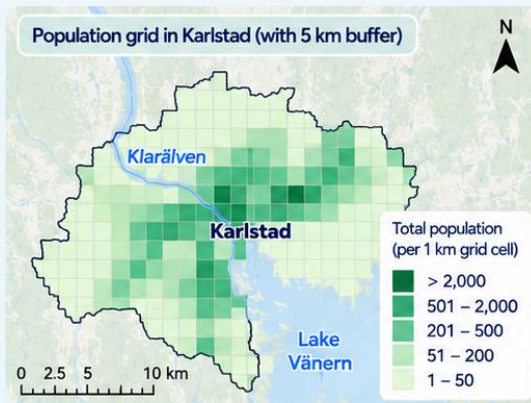
SCB Population Data

Gridded population data for Karlstad and surrounding areas



Statistiska centralbyrån
Statistics Sweden

Statistics Sweden (SCB) provides **open population data** on a regular grid, which can be used to estimate how many people may be affected by flooding and how many may lose access to critical infrastructure.



Dataset and content



SCB_population_Karlstad_5km.gpkg

1 × 1 km population grid
for Karlstad (with 5 km buffer)



Key attributes (examples)

- tot_pop – total population
- ald80_84 – 80–84 years
- ald85_89 – 85–89 years
- ald90_94 – 90–94 years
- ald95_99 – 95–99 years
- ald100w – 100+ years



Why is this dataset useful?



Total population

Estimate how many people live in areas that may be affected by flooding.



Vulnerable groups

Focus on elderly population (e.g. 80+ years) who may face higher risk.



Spatial analysis

Gridded data allows spatial aggregation and accessibility analysis (e.g. isochrones).



Steps to access the SCB population data

1

Go to the SCB open data portal

<https://www.scb.se/en/services/open-data/>



Statistiska centralbyrån

[Find statistics](#) [Services](#) [Documentation](#) [Leave feedback](#) [About SCB](#)

Open data

Download our open data and use it for analyses, research and visualizations.

2

Search for population grid data

Look for population statistics on a grid (1 × 1 km) for the desired area.



Population grid 1 × 1 km

Population by sex, age and area (1 × 1 km grid cells)

[GeoPackage](#)

[CSV](#)

[Statistics](#)

3

Select and download the dataset

Choose the relevant file for Karlstad and the surrounding area (e.g. 5 km buffer) and download the .gpkg file.

4

Use the dataset in the workshop

The file is already provided in our GitHub repository for easier access.



github.com/amaletaa/SESAC_RisCon26_Workshop



NMD2018 Land Cover Dataset

National land cover data for Sweden



The National Land Cover Database (NMD) is produced by the Swedish Environmental Protection Agency (Naturvårdsverket). It provides a consistent and detailed land cover map for the whole of Sweden, based on satellite data and other sources.



Dataset characteristics



Format

GeoTIFF (.tif)



Spatial resolution

10 m × 10 m



Coverage

Whole Sweden (national)



Content

Multiple thematic land cover classes



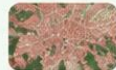
Main land cover classes (examples)



Forest



Agricultural areas



Urban and built-up areas



Water bodies



Grasslands



Wetlands and other



Steps to manually download land cover data

1

Go to the Naturvårdsverket website

<https://www.naturvardsverket.se/en/services-and-permits/maps-and-map-services/national-land-cover-database/>



MENU

TOPICS GUIDANCE LAWS AND REGULATIONS SERVICES AND PERMITS

Start > Services and permits > Maps and map services > National Land Cover Database (NMD)

National Land Cover Database (NMD)

2

Download the data

Under "Download the National Land Cover Database", select the ZIP file for the desired version (e.g. NMD2018).



S_lan_nmd2018bas_ogeneraliserad_v1_1.zip

3

Extract the GeoTIFF file

The ZIP file contains the GeoTIFF raster for Sweden.

4

Use for your study area

Clip the raster to your area of interest (e.g. Värmland County / Karlstad).



Tools & Python Libraries

Main libraries used in the workshop



osmnx

Download & query OpenStreetMap data

- Retrieve roads, buildings, POIs
- Create and analyze the road network
- Convert to a graph for routing



rasterio

Read / write raster files (GeoTIFF)

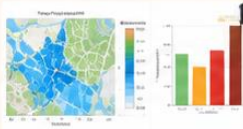
- Load MSB flood and land cover data
- Clip, resample, reproject
- Work with raster masks



matplotlib

Visualize maps and charts

- Create clear and informative plots
- Compare scenarios
- Customize maps and figures



numpy

Numerical operations on arrays

- Handle large raster and vector data
- Perform efficient calculations
- Work with arrays and matrices



geopandas

Manage vector geo-data (GeoDataFrames)

- Read, filter and manipulate OSM data
- Spatial analysis and joins
- Combine with other datasets



networkx

Graph analysis — road network routing

- Build network from OSM roads
- Calculate shortest routes
- Analyze accessibility under flooding



contextily

Web basemaps (CartoDB, OSM tiles)

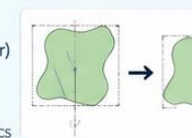
- Add background maps
- Create nice-looking maps
- Visualize results in a real-world context



shapely

Geometric operations (intersect, buffer)

- Clip data to study area
- Create buffers (e.g. around roads)
- Compute intersections and spatial metrics



Spatial Data Basics

How geographic information is represented and aligned

1 VECTOR

Points, Lines and Polygons

Vector data represent real-world features using **geometric shapes** with **attributes**.



Points
e.g. hospitals,
schools, fire stations

Lines
e.g. roads,
railways, rivers

Polygons
e.g. buildings,
land use,
administrative
boundaries

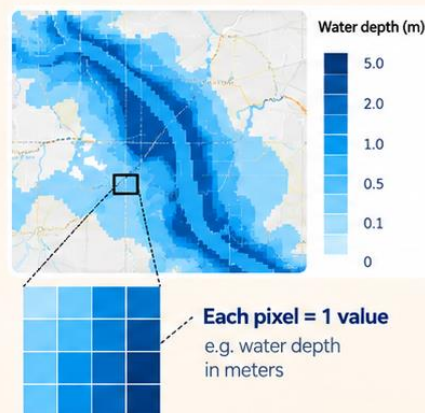
Used for in this workshop:

- Hospitals, schools, fire stations (points)
- Road network (lines)
- Administrative boundaries (polygons)

2 RASTER

Grid of pixels

Raster data consist of a grid of pixels, where each pixel stores a value.



Used for in this workshop:

- Flood depth (GeoTIFF) from MSB

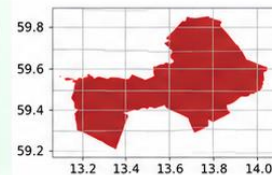
3 Coordinate Reference System (CRS)

Defines location and ensures alignment

A CRS defines how geographic locations are represented on the Earth's surface. Different datasets may use different coordinate systems, so we need to transform them to a common CRS for analysis.

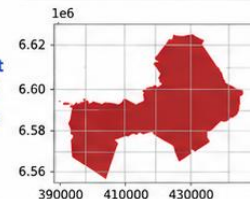
EPSG:4326
WGS84

Geographic coordinates
(latitude, longitude)



EPSG:3006
SWEREF99 TM

Projected coordinates
(meters)



Reproject
(to_crs)

Workshop CRS:



EPSG:3006 — SWEREF99 TM

We will reproject all datasets to this CRS before any spatial intersection or area calculation.



Before spatial analysis: make sure all datasets use the **same CRS**.

From Road Network to Accessibility

Using a graph to compute travel times and isochrones

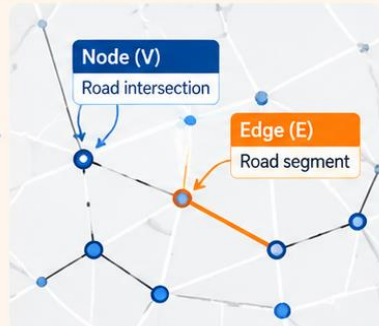
1 Road network

Roads from OpenStreetMap represent the transportation network in the study area.



2 Graph representation

The road network is modelled as a graph, where road intersections are **nodes** and road segments are **edges**.



- Node (V) – road intersection
- Edge (E) – road segment
- Direction – one-way / two-way

3 Edge attributes & travel time

Each edge has attributes such as length, speed and direction. We compute travel time for each edge:

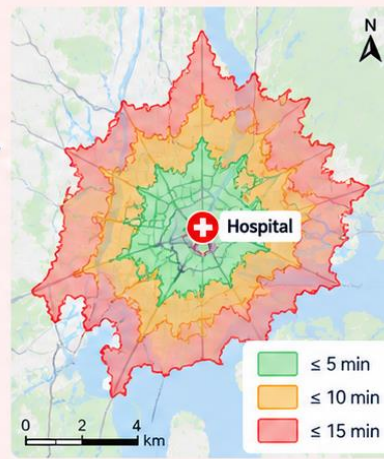
$$\text{Travel time [s]} = \frac{\text{Edge length [m]}}{\text{Speed [m/s]}}$$



- Length (m)
- Speed (km/h)
- Direction (one-way / two-way)
- Travel time (seconds)

4 Isochrones (accessibility)

An isochrone is the area that can be reached from a location within a given travel time along the road network.



Isochrone = area reachable from a facility within a given travel time along the road network.

Getting Started: Access the Workshop Notebooks

We will use Google Colab to run the provided notebooks (hosted on GitHub)

1

Open the workshop repository

Scan the QR code or go to the link to access all notebooks and data files.



https://github.com/amaletaa/SESAC_RisCon26_Workshop

amaletaa / SESAC_RisCon26_Workshop

notebooks/

data/

figures/

README.md



Workshop GitHub Repository

Scan the QR code or click the link to access all workshop notebooks, datasets and supporting files.



https://github.com/amaletaa/SESAC_RisCon26_Workshop



2

Open the notebook in Colab

In the repository, go to the notebooks folder and click on the notebook you want to open. Then click "Open in Colab".

02_network_analysis.ipynb



Open in Colab



3

Sign in to Google (if needed)

You need a Google account to use Colab. If you don't have one, create a free account at accounts.google.com.

Google

Sign in

Use your Google Account

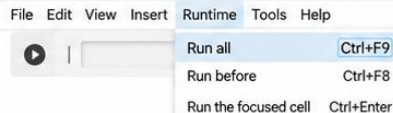
Email or phone

Next

4

Run the notebook

Once the notebook opens in Colab, click Runtime → Run all (or press Ctrl+F9) to run all cells and set up the environment.



5

Explore and experiment

The first run will download the required data and install the necessary packages (this may take ~2 minutes). After that, you can modify parameters and explore the results!



No software installation needed!

Google Colab runs entirely in your web browser, so you can use the workshop notebooks without installing any software on your computer.



Don't have a Google account?

You need a (free) Google account to use Colab. Create one at accounts.google.com if needed.



Tip

- The first run downloads data and installs packages (takes ~2 minutes).
- After that, you can re-run cells and modify parameters.
- All files (notebooks, data, instructions) are available in the GitHub repository.

THANK YOU

From flood hazard to societal impact

SESAC • AUTUMN 2026



SEPTEMBER
24–25 SEP

LUND

Beginner Earth Observation Analysis Series

A beginner-friendly introduction to Earth observation, GIS, and spatial analysis through lectures, demonstrations and hands-on exercises.

Free participation • No prior EO, GIS or programming experience required

REGISTER BY 12 SEP



OCTOBER
8 OCT

KARLSTAD

RisCon26: Flood & Accessibility Workshop

A hands-on introduction to analysing how flooding disrupts road networks and access to hospitals and other critical services.

13:00–14:30 • Beginner-friendly •
No programming experience required

RISCON26 REG. DEADLINE: 7 SEP

WORKSHOP-ONLY REG. DEADLINE: 11 SEP



NOVEMBER
19–20 NOV

LUND

Society from Space Challenge 2026

A two-day interdisciplinary hackathon connecting Earth Observation data science and social science to real-world urban challenges.

Two challenge tracks • Scania, Sweden –
Nairobi, Kenya • Free for selected participants

APPLY BY 30 SEP



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